Methodology: GPU DNS of Fractal-Interface Mixing in a Developing Duct

TorChannel — MERGE fractal-mixing study

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Abstract

We describe the numerical methodology used to test whether a fractal (Koch) scalar interface enhances mixing in a laminar developing pipe flow, and how the Schmidt number Sc is swept to probe the high-Peclet regime. The pipeline couples a GPU-accelerated, double-precision direct numerical simulation (DNS) of the incompressible Navier–Stokes equations — staggered grid, fractional-step pressure projection, IMEX time integration, Brinkman-penalized immersed pipe wall — to a conservative passive-scalar transport solver. To make the high-Sc, high-fractal-generation sweep tractable we exploit three physical properties of this configuration (parallel fully-developed flow, high-Peclet parabolicity, and an N-independent wall) that collapse the cost from $O(Pe)$ pseudo-time steps per case to a single downstream sweep on a frozen, once-computed velocity field.

1 Problem setup

The configuration is a short, developing, laminar duct of square outer cross-section $L_y \times L_z = 1 \times 1$ and streamwise length L_x, inside which a circular pipe of radius $R = 0.42$ is imposed as an immersed solid. Two co-flowing streams of a passive scalar $c \in [0, 1]$ enter the pipe at the inlet, separated by an interface. In the *baffle* configuration the dividing interface is a *Koch fractal* of generation N prescribed at the inlet; the pipe wall itself remains smooth and circular. The question is whether increasing the interfacial generation N — which multiplies the interfacial *area* without changing the 50/50 volume split — accelerates the downstream homogenisation of c, and how that effect scales with the Schmidt number

$$Sc = \frac{\nu}{D}, \qquad Pe = Re\, Sc = \frac{U_b L}{D}, \tag{1}$$

where ν is the kinematic viscosity, D the scalar diffusivity, U_b the bulk velocity and L a reference length. The molecular regime of interest (aqueous solutes) is $Sc \sim 10^3$, i.e. Pe up to $\sim 10^4$ at the chosen $Re = 40$.

2 Governing equations

The flow obeys the incompressible Navier–Stokes equations, with the passive scalar advected by the resulting divergence-free velocity:

$$\nabla \cdot \boldsymbol{u} = 0, \tag{2}$$

$$\frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = -\nabla p + \nu\, \nabla^2 \boldsymbol{u} + \boldsymbol{f}_{\mathrm{ib}}, \tag{3}$$

$$\frac{\partial c}{\partial t} + \boldsymbol{u} \cdot \nabla c = D\, \nabla^2 c, \qquad D = \frac{\nu}{Sc}. \tag{4}$$

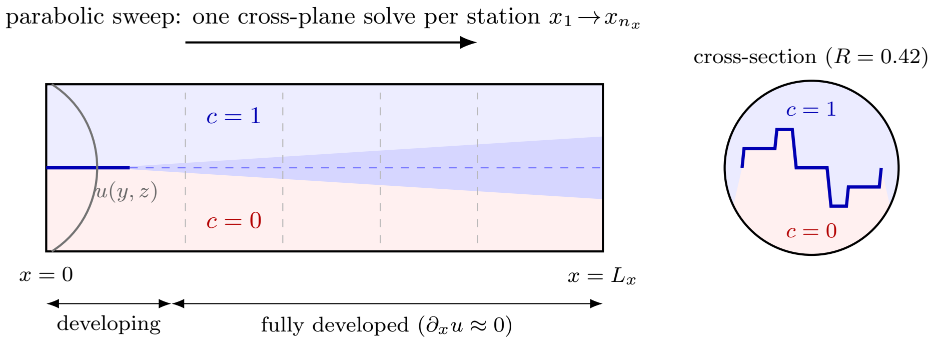


Figure 1: **(Sketch, schematic.)** Configuration. Two co-flowing scalar streams enter an immersed circular pipe separated by a Koch-folded interface (inset, cross-section). The fully-developed velocity is frozen and the steady scalar is obtained by a single downstream parabolic sweep.

All fields are stored in double precision throughout. The scalar is *passive*: it does not feed back into the momentum equation, so (4) carries no projection step. $\boldsymbol{f}_{\text{ib}}$ is the immersed-boundary penalization force (Section 5). Because the conservative flux form $\nabla\cdot(\boldsymbol{u}c)$ equals $\boldsymbol{u}\cdot\nabla c$ for a divergence-free field, advection is discretised in flux form to conserve $\langle c\rangle$ exactly.

3 Spatial discretisation

Staggered MAC grid. Velocities live on cell faces (the streamwise u on x-faces, v on y-faces, w on z-faces) while pressure and the scalar c are cell-centred. This Marker-and-Cell arrangement avoids odd–even pressure decoupling and gives a compact, exactly conservative divergence/gradient pair. All fields carry a single layer of ghost cells (indices 0 and -1 in each direction) through which boundary conditions are imposed.

Grid. The mesh is uniform in the periodic-capable x and y directions ($\mathrm{d}x = L_x/n_x$, $\mathrm{d}y = L_y/n_y$) and *stretched* in the wall-normal z via a hyperbolic-tangent mapping, clustering points near the walls; the variable spacings $\mathrm{d}z_c$ (centre-to-centre) and $\mathrm{d}z_f$ (face-to-face) enter every z-derivative. Spatial operators are second-order: central differences for diffusion, and conservative flux differences for advection. Performance-critical kernels are fused for the GPU.

4 Time integration: fractional-step projection

Momentum is advanced by a semi-implicit (IMEX) fractional-step (projection) method. Within one step of size Δt:

1. **Explicit terms (Adams–Bashforth 2).** Advection and the in-plane (x, y) diffusion are evaluated explicitly and combined with second-order Adams–Bashforth,

$$\boldsymbol{u}^* = \boldsymbol{u}^n + \Delta t\big(\tfrac{3}{2}\,\boldsymbol{R}^n - \tfrac{1}{2}\,\boldsymbol{R}^{n-1}\big), \quad \boldsymbol{R} = \nu\nabla^2_{xy}\boldsymbol{u} - (\boldsymbol{u}\cdot\nabla)\boldsymbol{u} + \boldsymbol{f}, \tag{5}$$

(the first step falls back to forward Euler).

2. **Implicit wall-normal diffusion (θ-method).** The stiff z-diffusion is treated implicitly, $\big(I - \theta\,\Delta t\,\nu\,\partial_{zz}\big)\boldsymbol{u}^{**} = \boldsymbol{u}^* + (1-\theta)\Delta t\,\nu\,\partial_{zz}\boldsymbol{u}^*$, with $\theta = \frac{1}{2}$ (Crank–Nicolson). On the stretched grid this is a tridiagonal system per wall-normal column, solved by a batched Thomas solver over all columns at once. Treating only the z-diffusion implicitly removes the most restrictive viscous stability limit while keeping the linear solve one-dimensional.

3. **Immersed-boundary penalization** (Section 5) is applied to $\boldsymbol{u}^{**}$ before projection.
4. **Pressure projection.** A Poisson equation enforces incompressibility,

$$\nabla^2 p = \frac{1}{\Delta t}\,\nabla\cdot \boldsymbol{u}^{**}, \qquad \boldsymbol{u}^{n+1} = \boldsymbol{u}^{**} - \Delta t\,\nabla p, \tag{6}$$

solved by an FFT-based fast Poisson solver in the periodic x, y planes combined with a tridiagonal solve in z (a direct banded solver is available as a fallback). The resulting $\boldsymbol{u}^{n+1}$ is discretely divergence-free.

5 Immersed pipe wall (Brinkman penalization)

The circular pipe is embedded in the Cartesian box by an implicit Brinkman (volume-penalization) force that drives the velocity to zero inside the solid:

$$\boldsymbol{f}_{\mathrm{ib}} = -\frac{\chi}{\eta}\,\boldsymbol{u}, \qquad \chi(\boldsymbol{x}) = \begin{cases} 1 & \text{solid (outside the pipe)} \\ 0 & \text{fluid (inside the pipe)} \end{cases} \tag{7}$$

with a small permeability $\eta = 10^{-4}$. The penalization is applied implicitly to the intermediate velocity *before* the projection, $\boldsymbol{u} \leftarrow \boldsymbol{u}/(1 + \Delta t\,\chi/\eta)$, so the fast FFT-Poisson solver is left untouched. The solid mask χ is sampled separately on each staggered face (χ_u, χ_v, χ_w) and at cell centres (χ_c); here it is the complement of the inscribed disc of radius $R = 0.42$ centred on the cross-section, giving a solid fraction ≈ 0.45. The bulk-flux controller and all mixing diagnostics are restricted to fluid cells ($\chi_c < \frac{1}{2}$) so that scalar diffusing into the (non-physical) solid is never counted as mixing.

6 Inflow / outflow boundary conditions

The duct is open (`bc_x = inout`): periodic in y only where specified, walls otherwise, and a prescribed inlet / convective outlet in x. The inlet streamwise velocity is a smooth duct profile, the product of parabolae in y and z that vanish at the walls,

$$u_{\mathrm{in}}(y,z) \propto \frac{4y(L_y - y)}{L_y^2}\,\frac{4z(L_z - z)}{L_z^2}, \tag{8}$$

masked to zero inside the solid and normalised so the fluid-area-mean inlet velocity equals U_b. The outflow imposes zero streamwise gradient with a global mass-flux correction enforcing $\int u_{\mathrm{out}} = \int u_{\mathrm{in}}$ over the fluid face, so the discrete domain conserves mass. The scalar inlet ghost is pinned to the prescribed interface $c_{\mathrm{inlet}}(y,z)$ each step; the scalar outflow is zero-gradient. Lateral walls use no-flux ($\partial_n c = 0$) for the scalar.

7 The baffle: a Koch-fractal inlet interface

In the *baffle* mode the pipe wall is smooth; the only N-dependent ingredient is the *scalar interface* prescribed at the inlet. The two streams meet on a curve that runs wall-to-wall across the pipe and is Koch-folded in the y-direction at generation N.

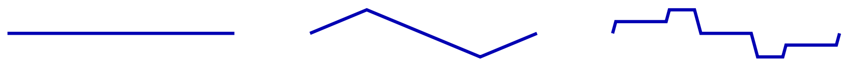
Generator. The interface is built from an area-balanced four-segment zigzag motif, each segment of length $1/r$:

$$(0,0) \to (\tfrac{1}{4}, a) \to (\tfrac{1}{2}, 0) \to (\tfrac{3}{4}, -a) \to (1,0), \qquad a = \sqrt{(1/r)^2 - (1/4)^2},$$

valid for $1 < r < 4$. Each outward triangle is matched by an equal inward indentation, so the fold preserves the exact 50/50 area split at every generation (the mean scalar is held at $\langle c \rangle = \frac{1}{2}$). Replacing every edge by an affine copy of the motif N times yields a generation-N polyline with fractal dimension

$$D_f = \frac{\log 4}{\log r}. \tag{9}$$

The interfacial length (hence the area available for diffusive flux) grows as $\sim (4/r)^N$ relative to the flat $N = 0$ baseline, while the enclosed area is unchanged — isolating the effect of interfacial area on mixing.



$N = 0$ $\qquad$ $N = 1$ $\qquad$ $N = 2$

Figure 2: Area-balanced Koch generator ($r = 3$, $D_f = \log 4/\log 3 \approx 1.26$). Each edge is replaced by the four-segment motif, multiplying interfacial length by $\sim 4/r$ per generation while preserving the 50/50 split. The inlet interface uses generation N of this fold; these are the actual polylines imposed.

Rasterisation. The cross-sectional field $c(y, z)$ is set from the signed distance ϕ to the open Koch polyline, smoothed over a prescribed interface thickness ε (a fixed number of cells):

$$c(y, z) = \tfrac{1}{2}\big[1 + \tanh(\phi/\varepsilon)\big]. \tag{10}$$

The distance *magnitude* comes from the open polyline, but its *sign* (which stream a cell belongs to) is taken from an even–odd point-in-polygon test against the interface closed along the far wall. This is robust at high N, where a naive nearest-segment side test produces wrong-sign corner artefacts ($N \geq 3$). The interface is homogeneous in x at the inlet; all streamwise structure that develops downstream is produced by the flow. A companion *surface_baffle* mode additionally folds the *pipe wall* into a Koch surface, differing from *baffle* only by the wall, for a controlled isolation of the surface-roughness effect.

8 Passive-scalar transport

The scalar lives at cell centres and is advanced with the same IMEX split as the momentum equations (AB2 for advection and in-plane diffusion; θ-method for wall-normal diffusion), reusing the batched tridiagonal solver. Advection is in conservative flux form,

$$\nabla \cdot (\boldsymbol{u}c) = \frac{F^x_{i+\frac{1}{2}} - F^x_{i-\frac{1}{2}}}{\mathrm{d}x} + \frac{F^y_{j+\frac{1}{2}} - F^y_{j-\frac{1}{2}}}{\mathrm{d}y} + \frac{F^z_{k+\frac{1}{2}} - F^z_{k-\frac{1}{2}}}{\mathrm{d}z_f}, \tag{11}$$

with face fluxes $F = U_{\text{face}}\, c_{\text{face}}$ built from the MAC face velocities. Two face reconstructions are available:

- **Central** (c_{face} = linear average) — second-order, used at moderate cell-Peclet.
- **van Leer TVD** — a MUSCL upwind reconstruction with the van Leer limiter, $c_{\text{face}} = c_{\text{up}} + \frac{1}{2}\,\psi$, monotone (no dispersive over/undershoot) at the high cell-Peclet of high-Sc runs. Since the face velocity is identically zero at every wall, the limiter never alters a wall flux.

The wall-normal boundary condition is no-flux (Neumann, $\partial_z c = 0$), which conserves the scalar mean and lets the segregation variance decay to zero — the correct closure for a mixing / decay study. A Dirichlet ($c = 0$) option exists for future sustained-gradient setups.

9 Mixing diagnostics

Mixing is quantified by the *intensity of segregation*, the fluid-volume-weighted, variance-based mixedness

$$M = \frac{\sigma_c}{\sigma_{\max}}, \qquad \sigma_c^2 = \langle (c - \bar{c})^2 \rangle, \qquad \sigma_{\max} = \sqrt{\bar{c}\,(1 - \bar{c})}, \tag{12}$$

where $\langle \cdot \rangle$ is the z-stretch-weighted average over fluid cells only. $M = 1$ is fully segregated (two pure streams); $M \to 0$ is fully mixed. For the developing duct the relevant object is the per-slice profile $M(x)$, evaluated plane-by-plane, from which a *mixing length* L_{mix} is read off as the streamwise station where $M(x)$ first falls below a threshold (e.g. $M = 0.9$), linearly interpolated between planes. The fractal effect is reported as the ratio $L_{\text{mix}}(N)/L_{\text{mix}}(0)$. A complementary scalar-dissipation diagnostic $\langle |\nabla c|^2 \rangle$ is also available; unlike the gravest-mode-dominated variance, it is directly sensitive to interfacial area and so captures the fractal's contribution that the variance metric alone can miss.

10 Acceleration: from $O(Pe)$ to one sweep

A direct time-march of (4) to steady state costs $\sim Pe$ pseudo-time steps and, at $Sc = 10^3$ on a fine cross-section, is infeasible. Three properties of this configuration collapse the cost.

10.1 Parabolic space-marching (high-Peclet) scalar solver

In a developing duct with $u > 0$ everywhere (no streamwise recirculation), the steady scalar balance

$$u\,\frac{\partial c}{\partial x} = D\left(\frac{\partial^2 c}{\partial y^2} + \frac{\partial^2 c}{\partial z^2}\right) - \left(v\,\frac{\partial c}{\partial y} + w\,\frac{\partial c}{\partial z}\right) \tag{13}$$

is obtained after *dropping the streamwise diffusion* $\partial_{xx}c$ — the *parabolic* (boundary-layer) approximation, asymptotically exact as $Pe \to \infty$ and, unlike a time-march, free of streamwise *numerical* diffusion. The equation is then parabolic in x and the steady field follows from a *single downstream sweep* — one cross-plane solve per x-station — instead of marching the unsteady field to steady state. The streamwise term uses second-order upwind (BDF2) from the two already-solved upstream planes; each cross-plane (y, z) diffusion solve uses alternating-direction implicit (ADI) line relaxation (a z-implicit then a y-implicit tridiagonal sweep), which converges in a handful of inner iterations — fastest precisely at high Sc, where the streamwise term dominates the diagonal. Cost is $O(n_x)$ plane solves rather than $O(Pe)$ full-field updates. For the smooth baffle $v = w = 0$, so transverse advection is switched off entirely.

10.2 Frozen, fully-developed velocity (parallel-flow assumption)

The scalar marcher runs on a *frozen* velocity field. The pipe wall is straight and x-independent, so the laminar flow becomes *fully developed* a short distance past the inlet: beyond the entrance length the profile $u(y, z)$ stops changing with x ($\partial_x u \approx 0$) and every downstream plane is identical. We verified this directly on the computed field — the cross-sectional profile is within 0.1% of its developed shape by $x \approx 2.8$ and $\partial_x u \sim 5 \times 10^{-5}$ thereafter. Consequently the velocity need only be solved over the short developing region, and extending it to an arbitrarily long scalar domain is exact: the marcher edge-clamps the streamwise interpolation to the developed profile, i.e. it repeats the one converged $u(y, z)$ downstream. This is the same parallel-flow premise that justifies dropping $\partial_{xx}c$ in the first place; it holds because the baffle wall does not vary with x (it would *not* hold for a streamwise-varying cross-section).

10.3 N-independent velocity, solved coarse and reused

In the baffle mode the Koch generation N lives only in the inlet *scalar*; the wall — hence the entire velocity field — is identical for every N. The developed velocity is therefore solved *once* and cached, then reused across the whole N-sweep. Moreover the smooth-pipe flow is well resolved on a *coarse* cross-section, whereas only the scalar needs the fine grid to resolve the Koch striations without false diffusion. We exploit this by solving the velocity on a coarse mesh and tri-linearly interpolating it (zeroed inside the fine solid mask) onto the fine scalar grid. The full N-sweep thus costs *one* coarse velocity solve plus N cheap parabolic scalar sweeps, in place of one fine velocity solve and a high-Sc time-march *per* N.

11 The Schmidt-number sweep

A sweep fixes the geometry (R, L_x, grid) and the threshold, and varies the Koch generation $N = 0, 1, 2, \ldots$ at a target Schmidt number. For each N the parabolic marcher returns the steady $c(x, y, z)$ on the frozen developed velocity; from it we extract $M(x)$, the mid-plane and cross-section slices, and $L_{\mathrm{mix}}(N)$. Comparing $L_{\mathrm{mix}}(N)/L_{\mathrm{mix}}(0)$ across N, at increasing Sc (equivalently $Pe = Re\,Sc$), tests whether the fractal interface's extra area shortens the mixing length and whether that advantage persists into the molecular ($Sc \sim 10^3$) regime.

Validation. The scalar solver is checked against an analytic error-function diffusion profile, exact mean conservation, and zero numerical streamwise diffusion; the DNS divergence is monitored every step and the run aborts if it grows.

12 Preliminary results ($Sc = 10$, 100 and 1000)

Table 1 and Figure 3 collect the mixing length L_{mix} (the station where $M(x)$ first crosses the threshold) for the baffle, at the three Schmidt numbers run. The $Sc = 100$ and $Sc = 1000$ data come from the parabolic marcher; the $Sc = 10$ data from a direct time-march to steady state (an independent cross-check of the marcher). Because L_{mix} grows with Pe, the $Sc = 1000$ sweep uses a longer domain ($L_x = 18$ vs. 6); even so the field does *not* reach $M = 0.9$ within the domain, so for that case the threshold is raised to $M = 0.95$ (which all N do cross) until longer-domain runs are available. The ratio $L_{\mathrm{mix}}(N)/L_{\mathrm{mix}}(0)$ is nearly threshold-independent, so the curves remain directly comparable.

	$Sc = 10$ (M=0.9)		$Sc = 100$ (M=0.9)		$Sc = 1000$ (M=0.95)	
N	L_{mix}	ratio	L_{mix}	ratio	L_{mix}	ratio
0	0.492	1.000	5.392	1.000	14.273	1.000
1	0.303	0.616	3.821	0.709	7.856	0.550
2	0.198	0.402	2.358	0.437	5.934	0.416
3	0.152	0.309	1.975	0.366	4.057	0.284
4	0.136	0.276	1.914	0.355	3.571	0.250
5	—	—	—	—	3.496	0.245

Table 1: Mixing length L_{mix} (station where $M(x)$ crosses the threshold) and its ratio to the flat $N = 0$ baseline, baffle mode, $R = 0.42$. $Sc = 10, 100$ use $L_x = 6$ and $M = 0.9$; $Sc = 1000$ uses $L_x = 18$ and a raised $M = 0.95$ (the Pe is too high to reach $M = 0.9$ within the domain — a longer-domain sweep is planned).

The mechanism is visible directly in the cross-sections (Figure 4): higher N injects more interfacial length into the same disc, and the steep transverse scalar gradients it creates are

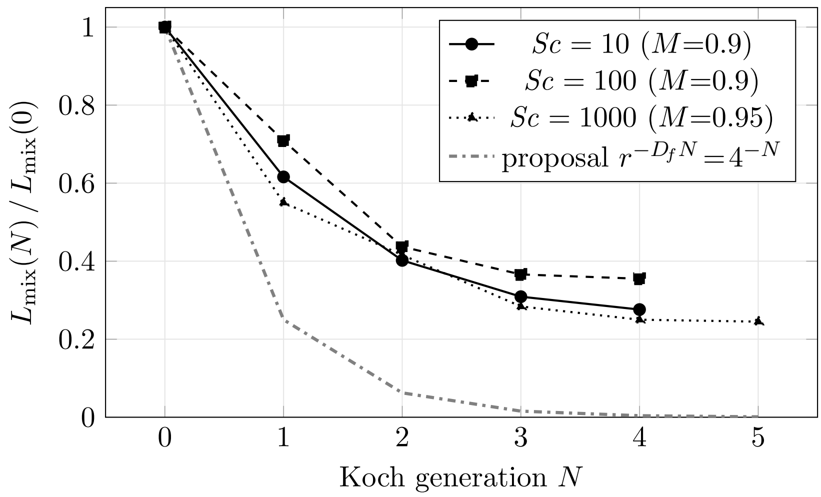


Figure 3: Fractal mixing enhancement: the mixing length falls steeply with generation N. The enhancement *strengthens with* Sc — the $Sc = 1000$ ratio reaches 0.25 by $N = 5$ — and the onset of saturation moves to higher N as Sc grows (the $Sc = 10$ curve is flat by $N = 3$, while $Sc = 1000$ still improves from $N = 4$ to $N = 5$). Ratios are nearly threshold-independent. The grey dash-dotted line is the proposal's diffusive-limit prediction $r^{-D_f N} = 4^{-N}$ (Eq. 4): the measured ratios decrease with N but saturate well above it, i.e. the finite-Pe duct realises part — not all — of the ideal exponential cut, and more of it as Sc increases.

ground down by diffusion over a shorter streamwise distance. The companion $M(x)$ decay (Figure 5) shows the same ordering as a single scalar per station, for both $Sc = 100$ and $Sc = 1000$.

13 Implications

1. **The fractal interface robustly shortens the mixing length, and the effect grows with Sc.** Folding the inlet interface to $N = 3$ cuts L_{mix} to 0.31, 0.37 and 0.28 of the flat baseline at $Sc = 10$, 100, 1000 respectively; by $N = 5$ at $Sc = 1000$ it reaches 0.25 (a fourfold reduction) — with the same volume split and the same wall. This confirms, in a resolved laminar duct, the core MERGE premise that interfacial *area* (not bulk stirring) drives the enhancement, and that it *strengthens* toward the molecular ($Sc \sim 10^3$) regime of interest.

2. **Diminishing returns / a Schmidt-dependent optimal generation.** Most of the gain is in the first folds ($N\colon 0 \to 1 \to 2$); the curve then flattens. Folds finer than the local diffusion length are smeared by diffusion before they can laminate, so there is a finite optimal generation N^* beyond which extra folding does not pay — and N^* *grows with* Sc: the $Sc = 10$ curve has largely flattened by $N = 3$–4, $Sc = 100$ still improves at $N = 4$, and $Sc = 1000$ still improves from $N = 4$ to $N = 5$. Higher Sc keeps finer folds effective, so the proposal regime should reward deeper fractals.

3. **L_{mix} scales steeply with Pe.** The $N = 0$ baseline grows $0.49 \to 5.4 \to >18$ (at $M = 0.9$; 14.3 at the raised $M = 0.95$) for $Sc = 10 \to 100 \to 1000$ — roughly a decade per decade of Sc. This is why a high-Sc run needs a long domain and many pseudo-time steps, and is the direct motivation for the parabolic-marcher / frozen-velocity acceleration of Section 10.

4. **Interfacial folding dominates over wall roughness.** The *surface_baffle* (Koch folds added to the pipe *wall*) gives a weaker reduction than the interface baffle at equal N (e.g.

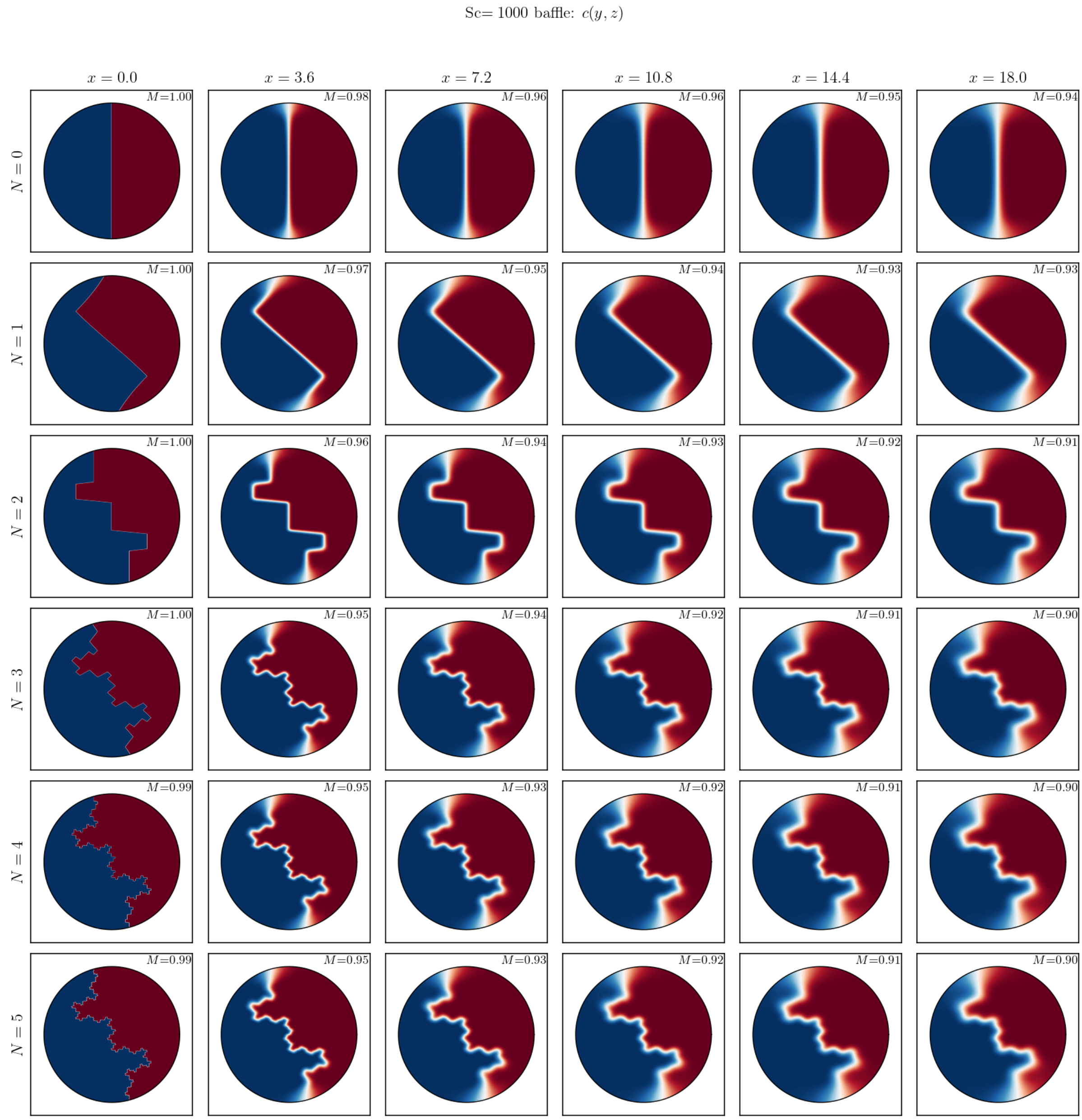


Figure 4: Steady scalar $c(y, z)$ in the pipe cross-section at $Sc = 1000$, from the parabolic marcher. Rows: Koch generation $N = 0$–5 (interfacial folding increases downward). Columns: streamwise station $x = 0$ (inlet) to $x = 18$ (outlet). The inlet interface ($x = 0$) is the prescribed generation-N Koch fold; downstream the fractal striations diffuse and the field homogenises faster at higher N (the per-panel M confirms the trend). Same 50/50 split and smooth wall for every N.

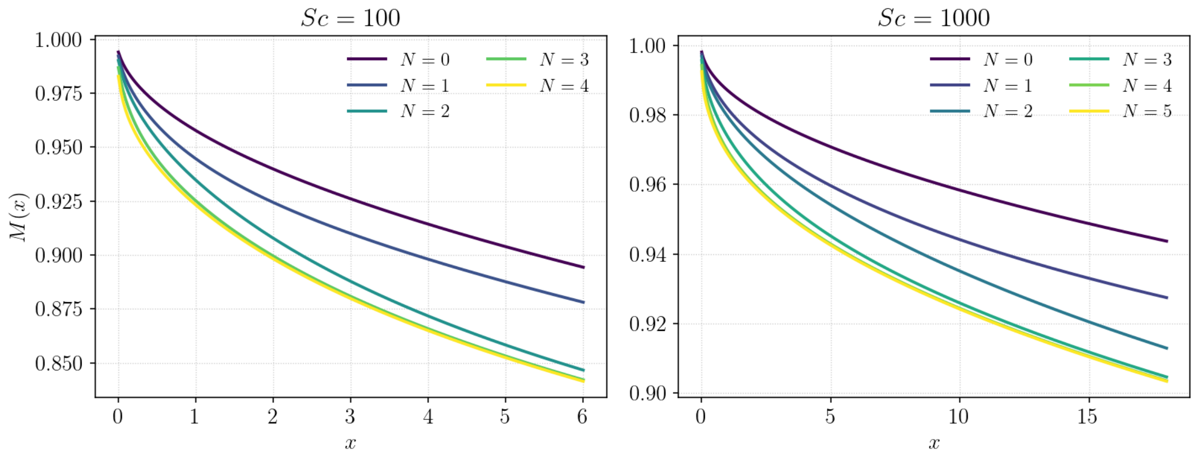


Figure 5: Segregation decay $M(x)$, generations N overlaid, for $Sc = 100$ ($L_x = 6$, left) and $Sc = 1000$ ($L_x = 18$, right). Higher N decays faster and further in both. At $Sc = 100$ the field crosses $M = 0.9$ within the domain; at $Sc = 1000$ it does not, motivating the raised $M = 0.95$ readout for the mixing length (Table 1).

at $Sc = 10$, $N = 3$: $L_{\text{mix}} = 0.21$ vs. 0.15), indicating that the dominant mechanism is lamination of the scalar interface, not corrugation-driven secondary flow.

5. **The acceleration is what makes the proposal regime reachable.** Because the velocity is fully developed and N-independent, the entire multi-N, high-Sc sweep collapses to one coarse velocity solve plus cheap parabolic scalar sweeps. Without it, true $Sc \sim 10^3$ by direct time-march would be infeasible; with it, the $Sc = 10^3$ sweep runs in minutes per generation.

Caveats. The $Sc = 10$ values come from a time-march (some cases stopped at the step cap, not fully steady) while $Sc = 100$ and $Sc = 1000$ use the parabolic marcher, so cross-Sc comparisons are indicative rather than exact. At $Sc = 1000$ the domain $L_x = 18$ is too short to reach $M = 0.9$, so L_{mix} there is reported at the *raised* threshold $M = 0.95$ (crossed by all N); a longer-domain sweep — planned — will let it be quoted at the same $M = 0.9$ as the other cases. The configuration and generator diagrams (Figs. 1, 2) are schematic; the quantitative figures (Figs. 3, 4, 5) are computed results.